COMPSCI701 Assignment 3 Report

Command Pattern	
Role	**Type(s)**
Invoker	**Game (Class):** In the *play( )* method, it reads user input, creates the correct Command, and calls appropriate *command.execute( )*. **Note:** the *execute( )* method is only called once inside the *play( )* method.
Receiver	**Game (Class) :** performs actions for *MoveCommand*, *NewGameCommand*, *SaveCommand*, *LoadGameCommand*, and *QuitCommand* via these methods - *handleMoveCommand( )*, *createNewGame( )*, *saveGame( )*, *loadGame( )*, and *requestQuit( )* respectively. **Note:** Also **TurnManager (Class) - Receiver for Move**: performs move actions via the method *executeMoveInTurn( )* once *Game.handleMoveCommand( )* delegates.
Command	**Command (Interface):** with the method *execute( )*.
Concrete Command(s)	**MoveCommand (Class - Concrete Class)** **NewGameCommand (Class - Concrete Class)** **SaveGameCommand (Class - Concrete Class)** **LoadGameCommand (Class - Concrete Class)** **QuitCommand (Class - Concrete Class)**
Client	**Game (Class):** acts as the factory of commands with the method *createCommand( )*.

Memento Pattern	
Role	**Type(s)**
Originator	**Game (Class):** . Encapsulates the game state (the Board and current turn state). . Creates *BoardSnapShot (snapShot)* in saveGame( ) and restores *BoardSnapShot (lastSnap)* in *loadGame( )*.
Memento	**BoardSnapShot (Class):** Stores the internal state of the Originator without exposing its implementation - capturing the amount of seeds in P1 and P2's houses and stores, and the currentPlayer at the turn the game is saved.
Caretaker	**GameCareTaker (Class):** Responsible for storing the Memento (the latest snapshot) or giving it back on load but never modifying them. Also, cleared on a new game.

Departures from Maintainability Advice	
Advice Identification	**Justification (50 words max)**

This is just from PMD Report: Game.java - "A value of 7 may denote a high amount of coupling within the class (threshold: 5)".	Game orchestrates UI, rules and design patterns, so references to *Board*, *TurnManager*, *GameResult*, *BoardSnapShot*, *GameCareTaker*, *Command* and *IO* are intentional. I considered decoupling (move *GameResult* into *TurnManager*; push save/load into *GameCareTaker*), but these either overload *TurnManager*'s responsibilities or blur Memento roles. Therefore coupling = 7 is a reasonable trade-off.